

1. Dataset loader
cases.jsonl (20 synthetic + 60 OncQA)



2. Perturbation engine
Name-Bank $\times$ Geo substitution, SHA-256 keyed



3. Model harness
OpenAI + Groq + AWS Bedrock, stdlib HTTP / boto3



4. LLM-as-annotator
Llama-3.1-8B, JSON-constrained, three-layer fallback



5. Metrics and statistics
TSR, RCR, RCER, GDI; Wilcoxon; BCa bootstrap

Cross-cutting components

- Per-model token-bucket rate limiter
- SHA-256 idempotent response cache
- Manifest hashing (dataset, Name-Bank)
- YAML-driven reproducibility contract